

Exponential Families

Inferential statistics is the science of learning from data. Data are typically viewed as random variables or vectors, but in contrast to our discussion of probability, distributions for these variables are generally unknown. In applications, it is often reasonable to assume that distributions come from a suitable class of distributions. In this chapter we introduce classes of distributions called *exponential families*. Examples include the binomial, Poisson, normal, exponential, geometric, and other distributions in regular use. From a theoretical perspective, exponential families are quite regular. In addition, moments for these distributions can often be computed easily using the differential identities in Section 2.4.

2.1 Densities and Parameters

Let μ be a measure on $\mathbb{R}^n$, let $h : \mathbb{R}^n \rightarrow \mathbb{R}$ be a nonnegative function, and let $T_1, \dots, T_s$ be measurable functions from $\mathbb{R}^n$ to $\mathbb{R}$. For $\eta \in \mathbb{R}^s$, define

$$A(\eta) = \log \int \exp \left[\sum_{i=1}^s \eta_i T_i(x) \right] h(x) d\mu(x). \quad (2.1)$$

Whenever $A(\eta) < \infty$, the function p_η given by

$$p_\eta(x) = \exp \left[\sum_{i=1}^s \eta_i T_i(x) - A(\eta) \right] h(x), \quad x \in \mathbb{R}^n, \quad (2.2)$$

integrates to one; that is, $\int p_\eta d\mu = 1$. So, this construction gives a family of probability densities indexed by η . The set

$$\Xi = \{\eta : A(\eta) < \infty\}$$

is called the *natural parameter space*, and the family of densities $\{p_\eta : \eta \in \Xi\}$ is called an *s-parameter exponential family in canonical form*.

Example 2.1. Suppose μ is Lebesgue measure on $\mathbb{R}$, $h = 1_{(0,\infty)}$, $s = 1$, and $T_1(x) = x$. Then

$$\begin{aligned} A(\eta) &= \log \int_0^\infty e^{\eta x} dx \\ &= \begin{cases} \log(-1/\eta), & \eta < 0; \\ \infty, & \eta \geq 0. \end{cases} \end{aligned}$$

Thus, $p_\eta(x) = \exp[\eta x - \log(-1/\eta)]1_{(0,\infty)}(x)$ is a density for $\eta \in \Xi = (-\infty, 0)$. In form, these are the exponential densities, which are usually parameterized by the mean or failure rate instead of the canonical parameter η here.

To allow other parameterizations for an exponential family of densities, let η be a function from some space Ω into Ξ and define

$$p_\theta(x) = \exp \left[\sum_{i=1}^s \eta_i(\theta) T_i(x) - B(\theta) \right] h(x)$$

for $\theta \in \Omega$, $x \in \mathbb{R}^n$, where $B(\theta) = A(\eta(\theta))$. Families $\{p_\theta : \theta \in \Omega\}$ of this form are called *s-parameter exponential families*.

Example 2.2. The normal distribution $N(\mu, \sigma^2)$ has density

$$\begin{aligned} p_\theta(x) &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)} \\ &= \frac{1}{\sqrt{2\pi}} \exp \left[\frac{\mu}{\sigma^2} x - \frac{1}{2\sigma^2} x^2 - \left(\frac{\mu^2}{2\sigma^2} + \log \sigma \right) \right], \end{aligned}$$

where $\theta = (\mu, \sigma^2)$. This is a two-parameter exponential family with $T_1(x) = x$, $T_2(x) = x^2$, $\eta_1(\theta) = \mu/\sigma^2$, $\eta_2(\theta) = -1/(2\sigma^2)$, $B(\theta) = \mu^2/(2\sigma^2) + \log \sigma$, and $h(x) = 1/\sqrt{2\pi}$.

Example 2.3. If $X_1, \dots, X_n$ is a random sample from $N(\mu, \sigma^2)$, then their joint density is

$$\begin{aligned} p_\theta(x_1, \dots, x_n) &= \prod_{i=1}^n \left[\frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x_i-\mu)^2/(2\sigma^2)} \right] \\ &= \frac{1}{(2\pi)^{n/2}} \exp \left[\frac{\mu}{\sigma^2} \sum_{i=1}^n x_i - \frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2 - n \left(\frac{\mu^2}{2\sigma^2} + \log \sigma \right) \right]. \end{aligned}$$

These densities also form a two-parameter exponential family with $T_1(x) = \sum_{i=1}^n x_i$, $T_2(x) = \sum_{i=1}^n x_i^2$, $\eta_1(\theta) = \mu/\sigma^2$, $\eta_2(\theta) = -1/(2\sigma^2)$, $B(\theta) = n[\mu^2/(2\sigma^2) + \log \sigma]$, and $h(x) = 1/(2\pi)^{n/2}$.

The similarity between these two examples is not accidental. If $X_1, \dots, X_n$ is a random sample with common marginal density

$$\exp \left[\sum_{i=1}^s \eta_i(\theta) T_i(x) - B(\theta) \right] h(x),$$

then their joint density is

$$\exp \left[\sum_{i=1}^s \eta_i(\theta) \left(\sum_{j=1}^n T_i(x_j) \right) - nB(\theta) \right] \prod_{j=1}^n h(x_j), \quad (2.3)$$

which is an s -parameter exponential family with the same functions $\eta_1, \dots, \eta_s$, and with

$$\tilde{T}_i(x) = \sum_{j=1}^n T_i(x_j), \quad \tilde{B}(\theta) = nB(\theta), \quad \tilde{h}(x) = \prod_{i=1}^n h(x_i),$$

where the tilde is used to indicate that the function is for the family of joint densities.

2.2 Differential Identities

In canonical exponential families it is possible to relate moments and cumulants for the statistics $T_1, \dots, T_s$ to derivatives of A . The following theorem plays a central role.

Theorem 2.4. *Let Ξ_f be the set of values for $\eta \in \mathbb{R}^s$ where*

$$\int |f(x)| \exp \left[\sum_{i=1}^s \eta_i T_i(x) \right] h(x) d\mu(x) < \infty.$$

Then the function

$$g(\eta) = \int f(x) \exp \left[\sum_{i=1}^s \eta_i T_i(x) \right] h(x) d\mu(x)$$

is continuous and has continuous partial derivatives of all orders for $\eta \in \Xi_f^\circ$ (the interior of Ξ_f). Furthermore, these derivatives can be computed by differentiation under the integral sign.

A proof of this result is given in Brown (1986), a monograph on exponential families with statistical applications. Although the proof is omitted here, key ideas from it are of independent interest and are presented in the next section. As an application of this result, if $f = 1$, then $\Xi_f = \Xi$, and, by (2.1),

$$g(\eta) = e^{A(\eta)} = \int \exp \left[\sum_{i=1}^s \eta_i T_i(x) \right] h(x) d\mu(x).$$

Differentiating this expression with respect to η_j , which can be done under the integral if $\eta \in \Xi^\circ$, gives

$$\begin{aligned} e^{A(\eta)} \frac{\partial A(\eta)}{\partial \eta_j} &= \int \frac{\partial}{\partial \eta_j} \exp \left[\sum_{i=1}^s \eta_i T_i(x) \right] h(x) d\mu(x) \\ &= \int T_j(x) \exp \left[\sum_{i=1}^s \eta_i T_i(x) \right] h(x) d\mu(x). \end{aligned}$$

Using the definition (2.2) of p_η , division by $e^{A(\eta)}$ gives

$$\frac{\partial A(\eta)}{\partial \eta_j} = \int T_j(x) p_\eta(x) d\mu(x).$$

This shows that if data X has density p_η with respect to μ , then

$$E_\eta T_j(X) = \frac{\partial A(\eta)}{\partial \eta_j} \quad (2.4)$$

for any $\eta \in \Xi^\circ$.

2.3 Dominated Convergence

When $s = 1$, (2.4) is obtained differentiating the identity

$$e^{A(\eta)} = \int e^{\eta T(x)} h(x) d\mu(x),$$

passing the derivative inside the integral. To understand why this should work, suppose the integral is finite for $\eta \in [-2\epsilon, 2\epsilon]$ and consider taking the derivative at $\eta = 0$. If the function is differentiable at zero, the derivative will be the following limit:

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{e^{A(\epsilon/n)} - e^{A(0)}}{\epsilon/n} &= \lim_{n \rightarrow \infty} \int \frac{e^{\epsilon T(x)/n} - 1}{\epsilon/n} h(x) d\mu(x) \\ &= \lim_{n \rightarrow \infty} \int f_n(x) d\mu(x), \end{aligned}$$

where

$$f_n(x) = \frac{e^{\epsilon T(x)/n} - 1}{\epsilon/n} h(x). \quad (2.5)$$

As $n \rightarrow \infty$, $f_n(x) \rightarrow f(x) \stackrel{\text{def}}{=} T(x)h(x)$. So the desired result follows provided

$$\int f_n d\mu \rightarrow \int f d\mu$$

as $n \rightarrow \infty$. This seems natural, but is not automatic (see Example 2.6). The following basic result gives a sufficient condition for this sort of convergence.

Theorem 2.5 (Dominated Convergence). *Let f_n , $n \geq 1$ be a sequence of functions with $|f_n| \leq g$ (a.e. μ) for all $n \geq 1$. If $\int g d\mu < \infty$ and $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ for a.e. x under μ , then*

$$\int f_n d\mu \rightarrow \int f d\mu$$

as $n \rightarrow \infty$.

Example 2.6. To appreciate the need for a “dominating” function g in this theorem, suppose μ is Lebesgue measure on $\mathbb{R}$, define $f_n = 1_{(n, n+1)}$, $n \geq 1$, and take $f = 0$. Then $f_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$, for all x . But $\int f_n d\mu = 1$, for all $n \geq 1$, and these values do not converge to $\int f d\mu = 0$.

To apply dominated convergence in our original example with f_n given by (2.5), the following bounds are useful:

$$|e^t - 1| \leq |t|e^{|t|}, \quad t \in \mathbb{R},$$

and

$$|t| \leq e^{|t|}, \quad t \in \mathbb{R}.$$

Using these,

$$\begin{aligned} \left| \frac{e^{\epsilon T(x)/n} - 1}{\epsilon/n} \right| &\leq \frac{|\epsilon T(x)/n|}{\epsilon/n} e^{|\epsilon T(x)/n|} \\ &\leq \frac{1}{\epsilon} |\epsilon T(x)| e^{|\epsilon T(x)|} \leq \frac{1}{\epsilon} e^{2\epsilon T(x)} \leq \frac{1}{\epsilon} \left(e^{2\epsilon T(x)} + e^{-2\epsilon T(x)} \right). \end{aligned}$$

The left-hand side of this bound multiplied by $h(x)$ is $|f_n(x)|$, so

$$|f_n(x)| \leq \frac{1}{\epsilon} \left(e^{2\epsilon T(x)} + e^{-2\epsilon T(x)} \right) h(x) \stackrel{\text{def}}{=} g(x).$$

The dominating function g has a finite integral because

$$\int e^{\pm 2\epsilon T(x)} h(x) d\mu(x) = e^{A(\pm 2\epsilon)} < \infty.$$

So, by dominated convergence $\int f_n d\mu \rightarrow \int f d\mu$ as $n \rightarrow \infty$, as desired.

2.4 Moments, Cumulants, and Generating Functions

Let $T = (T_1, \dots, T_s)'$ be a random vector in $\mathbb{R}^s$. Note that the dot product of T with a constant $u \in \mathbb{R}^s$ is $u \cdot T = u'T$. The *moment generating function* of T is defined as

$$M_T(u) = Ee^{u_1 T_1 + \dots + u_s T_s} = Ee^{u'T}, \quad u \in \mathbb{R}^s,$$

and the *cumulant generating function* is

$$K_T(u) = \log M_T(u).$$

According to the following lemma, the moment generating function M_X determines the distribution of X , at least if it is finite in some open interval.

Lemma 2.7. *If the moment generating functions $M_X(u)$ and $M_Y(u)$ for two random vectors X and Y are finite and agree for u in some set with a nonempty interior, then X and Y have the same distribution, $P_X = P_Y$.*

Expectations of products of powers of $T_1, \dots, T_s$ are called *moments* of T , denoted

$$\alpha_{r_1, \dots, r_s} = E[T_1^{r_1} \times \dots \times T_s^{r_s}].$$

The following result shows that these moments can generally be found by differentiating M_T at $u = 0$. The proof is omitted, but is similar to the proof of Theorem 2.4. Here, dominated convergence would be used to justify differentiation under an expectation.

Theorem 2.8. *If M_T is finite in some neighborhood of the origin, then M_T has continuous derivatives of all orders at the origin, and*

$$\alpha_{r_1, \dots, r_s} = \left. \frac{\partial^{r_1}}{\partial u_1^{r_1}} \cdots \frac{\partial^{r_s}}{\partial u_s^{r_s}} M_T(u) \right|_{u=0}.$$

The corresponding derivatives of K_T are called *cumulants*, denoted

$$\kappa_{r_1, \dots, r_s} = \left. \frac{\partial^{r_1}}{\partial u_1^{r_1}} \cdots \frac{\partial^{r_s}}{\partial u_s^{r_s}} K_T(u) \right|_{u=0}.$$

When $s = 1$, $K'_T = M'_T/M_T$ and $K''_T = [M_T M''_T - (M'_T)^2]/M_T^2$. At $u = 0$, these equations give

$$\kappa_1 = ET \quad \text{and} \quad \kappa_2 = ET^2 - (ET)^2 = \text{Var}(T).$$

Generating functions can be quite useful in the study of sums of independent random vectors. As a preliminary to this investigation, the following lemma shows that in regular situations, the expectation of a product of independent variables is the product of the expectations.

Lemma 2.9. *Suppose X and Y are independent random variables. If X and Y are both positive, or if $E|X|$ and $E|Y|$ are both finite, then*

$$E|XY| = E|X| \times E|Y|.$$

Proof. Viewing $|XY|$ as a function g of $Z = \begin{pmatrix} X \\ Y \end{pmatrix} \sim P_X \times P_Y$, by Fubini's theorem,

$$E|XY| = \int g d(P_X \times P_Y) = \int \left(\int |x||y| dP_X(x) \right) dP_Y(y).$$

The inner integral is $|y|E|X|$, and the outer integral then gives $E|X| \times E|Y|$, so $E|XY| = E|X| \times E|Y|$. This proves the lemma if X and Y are both positive, because then $X = |X|$ and $Y = |Y|$. If $E|X| < \infty$ and $E|Y| < \infty$, then $E|XY| < \infty$, so the same steps omitting absolute values prove the lemma. $\square$

By iteration, this lemma extends easily to products of several independent variables.

Suppose $T = Y_1 + \cdots + Y_n$, where $Y_1, \dots, Y_n$ are independent random vectors in $\mathbb{R}^s$. Then by Proposition 1.19, the random variables $e^{u \cdot Y_1}, \dots, e^{u \cdot Y_n}$ are independent, and

$$M_T(u) = Ee^{u \cdot T} = E[e^{u \cdot Y_1} \times \cdots \times e^{u \cdot Y_n}] = M_{Y_1}(u) \times \cdots \times M_{Y_n}(u).$$

Taking logarithms,

$$K_T(u) = K_{Y_1}(u) + \cdots + K_{Y_n}(u).$$

Derivatives at the origin give cumulants, and thus cumulants for the sum T will equal the sum of the corresponding cumulants of $Y_1, \dots, Y_n$. This is a well-known result for the mean and variance.

If X has density from a canonical exponential family (2.2), and if $T = T(X)$, then T has moment generating function

$$\begin{aligned} E_\eta e^{u \cdot T(X)} &= \int e^{u \cdot T(x)} e^{\eta \cdot T(x) - A(\eta)} h(x) d\mu(x) \\ &= e^{A(u+\eta) - A(\eta)} \int e^{(u+\eta) \cdot T(x) - A(u+\eta)} h(x) d\mu(x), \end{aligned}$$

provided $u + \eta \in \Xi$. The final integrand is $p_{u+\eta}$, which integrates to one. So, the moment generating function is $e^{A(u+\eta) - A(\eta)}$, and the cumulant generating function is

$$K_T(u) = A(u + \eta) - A(\eta).$$

Taking derivatives, the cumulants for T are

$$\kappa_{r_1, \dots, r_s} = \frac{\partial^{r_1}}{\partial \eta_1^{r_1}} \cdots \frac{\partial^{r_s}}{\partial \eta_s^{r_s}} A(\eta).$$

Example 2.10. If X has the Poisson distribution with mean λ , then

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!} = \frac{1}{x!} e^{x \log \lambda - \lambda}, \quad x = 0, 1, \dots$$

The mass functions for X form an exponential family, but the family is not in canonical form. The canonical parameter here is $\eta = \log \lambda$. The mass function expressed using η is

$$P(X = x) = \frac{1}{x!} \exp[\eta x - e^\eta], \quad x = 0, 1, \dots,$$

and so $A(\eta) = e^\eta$. Taking derivatives, all of the cumulants of $T = X$ are $e^\eta = \lambda$.

Example 2.11. The class of normal densities formed by varying μ with σ^2 fixed can be written as

$$p_\mu(x) = \exp\left[\frac{\mu x}{\sigma^2} - \frac{\mu^2}{2\sigma^2}\right] \frac{e^{-x^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}}.$$

These densities form an exponential family with $T(x) = x$, canonical parameter $\eta = \mu/\sigma^2$, and $A(\eta) = \sigma^2 \eta^2/2$. The first two cumulants are $\kappa_1 = A'(\eta) = \sigma^2 \eta = \mu$ and $\kappa_2 = A''(\eta) = \sigma^2$. Because A is quadratic, all higher-order cumulants, $\kappa_3, \kappa_4, \dots$, are zero.

To calculate moments from cumulants when $s = 1$, repeatedly differentiate the identity $M = e^K$. This gives $M' = K'e^K$, $M'' = (K'' + K'^2)e^K$, $M''' = (K''' + 3K'K'' + K'^3)e^K$, and $M'''' = (K'''' + 3K'''^2 + 4K'K'''' + 6K'^2K'' + K'^4)e^K$. At zero, these equations give

$$ET = \kappa_1, \quad ET^2 = \kappa_2 + \kappa_1^2, \quad ET^3 = \kappa_3 + 3\kappa_1\kappa_2 + \kappa_1^3,$$

and

$$ET^4 = \kappa_4 + 3\kappa_2^2 + 4\kappa_1\kappa_3 + 6\kappa_1^2\kappa_2 + \kappa_1^4.$$

For instance, if $X \sim \text{Poisson}(\lambda)$, $EX = \lambda$, $EX^2 = \lambda + \lambda^2$, $EX^3 = \lambda + 3\lambda^2 + \lambda^3$, and $EX^4 = \lambda + 7\lambda^2 + 6\lambda^3 + \lambda^4$, and if $X \sim N(\mu, \sigma^2)$, $EX^3 = 3\mu\sigma^2 + \mu^3$ and $EX^4 = 3\sigma^4 + 6\mu^2\sigma^2 + \mu^4$.

The expressions above expressing moments as functions of cumulants can be solved to express cumulants as functions of moments. The algebra is easier if the variables are centered. Note that for $c \in \mathbb{R}^s$,

$$M_{T+c}(u) = Ee^{u \cdot (T+c)} = e^{u \cdot c} Ee^{u \cdot T} = e^{u \cdot c} M_T(u),$$

and so $K_{T+c}(u) = u \cdot c + K_T(u)$. Taking derivatives, it is clear that the constant c only affects first-order cumulants. So with $s = 1$, if $j \geq 2$, the j th cumulant κ_j for T will be the same as the j th cumulant for $T - ET$. The equations above then give

$$\kappa_3 = E(T - ET)^3$$

and

$$E(T - ET)^4 = \kappa_4 + 3\kappa_2^2,$$

and so

$$\kappa_4 = E(T - ET)^4 - 3\text{Var}^2(T).$$

In higher dimensions, the first-order cumulants are the means of $T_1, \dots, T_s$, and second-order cumulants are covariances between these variables. Formulas for mixed cumulants in higher dimensions become quite complicated as the order increases.

2.5 Problems¹

- *1. Consider independent Bernoulli trials with success probability p and let X be the number of failures before the first success. Then $P(X = x) = p(1 - p)^x$, for $x = 0, 1, \dots$, and X has the geometric distribution with parameter p , introduced in Problem 1.17.
 - a) Show that the geometric distributions form an exponential family.
 - b) Write the densities for the family in canonical form, identifying the canonical parameter η , and the function $A(\eta)$.
 - c) Find the mean of the geometric distribution using a differential identity.
 - d) Suppose $X_1, \dots, X_n$ are i.i.d. from a geometric distribution. Show that the joint distributions form an exponential family, and find the mean and variance of T .
- *2. Determine the canonical parameter space Ξ , and find densities for the one-parameter exponential family with μ Lebesgue measure on $\mathbb{R}^2$, $h(x, y) = \exp[-(x^2 + y^2)/2]/(2\pi)$, and $T(x, y) = xy$.
3. Suppose that $X_1, \dots, X_n$ are independent random variables and that for $i = 1, \dots, n$, X_i has a Poisson distribution with mean $\lambda_i = \exp(\alpha + \beta t_i)$, where $t_1, \dots, t_n$ are observed constants and α and β are unknown parameters. Show that the joint distributions for $X_1, \dots, X_n$ form a two-parameter exponential family and identify the statistics T_1 and T_2 .
- *4. Find the natural parameter space Ξ and densities p_η for a canonical one-parameter exponential family with μ Lebesgue measure on $\mathbb{R}$, $T_1(x) = \log x$, and $h(x) = (1 - x)^2$, $x \in (0, 1)$, and $h(x) = 0$, $x \notin (0, 1)$.
- *5. Find the natural parameter space Ξ and densities p_η for a canonical one-parameter exponential family with μ Lebesgue measure on $\mathbb{R}$, $T_1(x) = -x$, and $h(x) = e^{-2\sqrt{x}}/\sqrt{x}$, $x > 0$, and $h(x) = 0$, $x \leq 0$. (Hint: After a change of variables, relevant integrals will look like integrals against a normal density. You should be able to express the answer using Φ , the standard normal cumulative distribution function.) Also, determine the mean and variance for a variable X with this density.

¹ Solutions to the starred problems are given at the back of the book.

- *6. Find the natural parameter space Ξ and densities p_η for a canonical two-parameter exponential family with μ counting measure on $\{0, 1, 2\}$, $T_1(x) = x$, $T_2(x) = x^2$, and $h(x) = 1$ for $x \in \{0, 1, 2\}$.
- *7. Suppose $X_1, \dots, X_n$ are independent geometric variables with p_i the success probability for X_i . Suppose these success probabilities are related to a sequence of “independent” variables $t_1, \dots, t_n$, viewed as known constants, through

$$p_i = 1 - \exp(\alpha + \beta t_i), \quad i = 1, \dots, n.$$

Show that the joint densities for $X_1, \dots, X_n$ form a two-parameter exponential family, and identify the statistics T_1 and T_2 .

- *8. Assume that $X_1, \dots, X_n$ are independent random variables with $X_i \sim N(\alpha + \beta t_i, 1)$, where $t_1, \dots, t_n$ are observed constants and α and β are unknown parameters. Show that the joint distributions for $X_1, \dots, X_n$ form a two-parameter exponential family, and identify the statistics T_1 and T_2 .
- *9. Suppose that $X_1, \dots, X_n$ are independent Bernoulli variables (a random variable is Bernoulli if it only takes on values 0 and 1) with

$$P(X_i = 1) = \frac{\exp(\alpha + \beta t_i)}{1 + \exp(\alpha + \beta t_i)}.$$

Show that the joint distributions for $X_1, \dots, X_n$ form a two-parameter exponential family, and identify the statistics T_1 and T_2 .

10. Suppose a researcher is interested in how the variance of a response Y depends on an independent variable x . Natural models might be those in which $Y_1, \dots, Y_n$ are independent mean zero normal variables with the variance of Y_i some function of a linear function of x_i :

$$\text{Var}(Y_i) = g(\theta_1 + \theta_2 x_i).$$

Suggest a form for the function g such that the joint distributions for the Y_i , as the parameters θ vary, form a two-parameter exponential family.

11. Find the natural parameter space Ξ and densities p_η for a canonical one-parameter exponential family with μ Lebesgue measure on $\mathbb{R}$, $T_1(x) = x$, and $h(x) = \sin x$, $x \in (0, \pi)$, and $h(x) = 0$, $x \notin (0, \pi)$.
12. *Truncation.* Let $\{p_\theta : \theta \in \Omega\}$ be an exponential family of densities with respect to some measure μ , where

$$p_\theta(x) = h(x) \exp \left[\sum_{i=1}^s \eta_i(\theta) T_i(x) - B(\theta) \right].$$

In some situations, a potential observation X with density p_θ can only be observed if it happens to lie in some region S . For regularity, assume that $\Lambda(\theta) \stackrel{\text{def}}{=} P_\theta(X \in S) > 0$. In this case, the appropriate distribution for the observed variable Y is given by

$$P_\theta(Y \in B) = P_\theta(X \in B | X \in S), \quad B \in \mathcal{B}.$$

This distribution for Y is called the *truncation* of the distribution for X to the set S .

- a) Show that Y has a density with respect to μ , giving a formula for its density q_θ .
 - b) Show that the densities q_θ , $\theta \in \Omega$, form an exponential family.
13. Find densities p_η for a canonical one-parameter exponential family if μ is counting measure on $\mathcal{X}_0 = \{-1, 0, 1\}^3$, h is identically one, and $T(x)$ is the median of x_1, x_2 , and x_3 .
- *15. For an exponential family in canonical form, $ET_j = \partial A(\eta)/\partial \eta_j$. This can be written in vector form as $ET = \nabla A(\eta)$. Derive an analogous differential formula for $E_\theta T$ for an s -parameter exponential family that is not in canonical form. Assume that Ω has dimension s . Hint: Differentiation under the integral sign should give a system of linear equations. Write these equations in matrix form.
16. Find the natural parameter space Ξ and densities p_η for a canonical one-parameter exponential family with μ counting measure on $\{1, 2, \dots\}$, $h(x) = x^2$, and $T(x) = -x$. Also, determine the mean and variance for a random variable X with this density. Hint: Consider what Theorem 2.4 has to say about derivatives of $\sum_{x=1}^\infty e^{-\eta x}$.
- *17. Let μ denote counting measure on $\{1, 2, \dots\}$. One common definition for $\sum_{k=1}^\infty f(k)$ is $\lim_{n \rightarrow \infty} \sum_{k=1}^n f(k)$, and another definition is $\int f d\mu$.
- a) Use the dominated convergence theorem to show that the two definitions give the same answer when $\int |f| d\mu < \infty$. Hint: Find functions f_n , $n = 1, 2, \dots$, so that $\sum_{k=1}^n f(k) = \int f_n d\mu$.
 - b) Use the monotone convergence theorem, given in Problem 1.25, to show the definitions agree if $f(k) \geq 0$ for all $k = 1, 2, \dots$.
 - c) Suppose $\lim_{n \rightarrow \infty} f(n) = 0$ and that $\int f^+ d\mu = \int f^- d\mu = \infty$ (so that $\int f d\mu$ is undefined). Let K be an arbitrary constant. Show that the list $f(1), f(2), \dots$ can be rearranged to form a new list $g(1), g(2), \dots$ so that

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n g(k) = K.$$

18. Let λ be Lebesgue measure on $(0, \infty)$. The “Riemann” definition of $\int_0^\infty f(x) dx$ for a continuous function f is

$$\lim_{c \rightarrow \infty} \int_0^c f(x) dx,$$

when the limit exists. Another definition is $\int f d\lambda$. Use the dominated convergence theorem to show that these definitions agree when f is integrable, $\int |f| d\lambda < \infty$. Hint: Let c_n be a sequence of constants with $c_n \rightarrow \infty$, and find functions f_n such that $\int f_n(x) dx = \int_0^{c_n} f(x) dx$.

- *19. Let p_n , $n = 1, 2, \dots$, and p be probability densities with respect to a measure μ , and let P_n , $n = 1, 2, \dots$, and P be the corresponding probability measures.

- a) Show that if $p_n(x) \rightarrow p(x)$ as $n \rightarrow \infty$, then $\int |p_n - p| d\mu \rightarrow 0$. Hint: First use the fact that $\int (p_n - p) d\mu = 0$ to argue that $\int |p_n - p| d\mu = 2 \int (p - p_n)^+ d\mu$. Then use dominated convergence.
- b) Show that $|P_n(A) - P(A)| \leq \int |p_n - p| d\mu$. Hint: Use indicators and the bound $|\int f d\mu| \leq \int |f| d\mu$.

Remark: Distributions P_n , $n \geq 1$, are said to *converge strongly* to P if $\sup_A |P_n(A) - P(A)| \rightarrow 0$. The two parts above show that pointwise convergence of p_n to p implies strong convergence. This was discovered by Scheffé.

20. Let h be a bounded differentiable function on $[0, \infty)$, vanishing at zero, $h(0) = 0$.

- a) Show that

$$\int_0^\infty |h(1/x^2)| dx < \infty.$$

Hint: Because h is differentiable at 0, $h(x)/x \rightarrow h'(0)$ as $x \downarrow 0$, and $|h(x)| \leq cx$ for x sufficiently small.

- b) If Z has a standard normal distribution, $Z \sim N(0, 1)$, find

$$\lim_{n \rightarrow \infty} n E h(1/(n^2 Z^2)).$$

Hint: Be careful with your argument: the answer should not be zero.

21. Let μ be counting measure on $\{1, 2, \dots\}$, and let $f_n = c_n 1_{\{n\}}$, $n = 1, 2, \dots$, for some constants $c_1, c_2, \dots$

- a) Find $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ for $x = 1, 2, \dots$
- b) Show that these functions f_n can be dominated by an integrable function; that is, there exists g with $\int g d\mu < \infty$ and $|f_n| \leq g$, $n = 1, 2, \dots$, if and only if

$$\sum_{n=1}^{\infty} |c_n| < \infty.$$

- c) Find constants $c_1, c_2, \dots$ that provide an example of functions f_n that cannot be dominated by an integrable function, so the assumption of the dominated convergence theorem fails, but $\int f_n d\mu \rightarrow \int f d\mu$.

- *22. Suppose X is absolutely continuous with density

$$p_\theta(x) = \begin{cases} \frac{e^{-(x-\theta)^2/2}}{\sqrt{2\pi}\Phi(\theta)}, & x > 0; \\ 0, & \text{otherwise.} \end{cases}$$

Find the moment generating function of X . Compute the mean and variance of X .

- *23. Suppose $Z \sim N(0, 1)$. Find the first four cumulants of Z^2 . Hint: Consider the exponential family $N(0, \sigma^2)$.
- *24. Find the first four cumulants of $T = XY$ when X and Y are independent standard normal variates.
- *25. Find the third and fourth cumulants of the geometric distribution.
- *26. Find the third cumulant and third moment of the binomial distribution with n trials and success probability p .
- *27. Let T be a random vector in $\mathbb{R}^2$.
 a) Express $\kappa_{2,1}$ as a function of the moments of T .
 b) Assume $ET_1 = ET_2 = 0$ and give an expression for $\kappa_{2,2}$ in terms of moments of T .
- *28. Suppose $X \sim \Gamma(\alpha, 1/\lambda)$, with density

$$\frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}, \quad x > 0.$$

Find the cumulants of $T = (X, \log X)$ of order 3 or less. The answer will involve $\psi(\alpha) = d \log \Gamma(\alpha) / d\alpha = \Gamma'(\alpha) / \Gamma(\alpha)$.

29. Let $X_1, \dots, X_n$ be independent random variables, and let α_i and t_i , $i = 1, \dots, n$, be known constants. Suppose $X_i \sim \Gamma(\alpha_i, 1/\lambda_i)$ with $\lambda_i = \theta_1 + \theta_2 t_i$, $i = 1, \dots, n$, where θ_1 and θ_2 are unknown parameters. Show that the joint distributions form a two-parameter exponential family. Identify the statistic T and give its mean and covariance matrix. (Similar models arise in “parameter design” experiments used to study the effects of various factors on process variation.)
30. In independent Bernoulli trials with success probability p , the variable X counting the number of failures before the m th success has a negative binomial distribution with mass function

$$P(X = x) = \binom{m+x-1}{m-1} p^m (1-p)^x, \quad x = 0, 1, \dots$$

Find the moment generating function of X , along with the first three moments and first three cumulants of X .

31. An estimator $\hat{\theta}$ is called *unbiased* for a parameter θ if $E\hat{\theta} = \theta$. If $X_1, \dots, X_n$ are i.i.d., then the sample moment

$$\hat{\alpha}_r = \frac{1}{n} \sum_{i=1}^n X_i^r$$

is an unbiased estimator of $\alpha_r = EX_i^r$. Unbiased estimators for cumulants are called *K-statistics*. They are a bit harder to identify than unbiased estimators for moments, because cumulants depend on powers of moments. For example, $\kappa_2 = \alpha_2 - \alpha_1^2$.

- a) One natural estimator for α_1^2 is $\overline{X}^2 = \hat{\alpha}_1^2$. Find the expected value of this estimator. When is it biased?

b) Show that

$$\binom{n}{2}^{-1} \sum_{1 \leq i < j \leq n} X_i X_j$$

is unbiased for α_1^2 . Give a formula relating this estimator to sample moments $\hat{\alpha}_1$ and $\hat{\alpha}_2$.

- c) Give an unbiased estimator for $\kappa_2 = \text{Var}(X_i)$ based on sample moments $\hat{\alpha}_1$ and $\hat{\alpha}_2$.
 d) Find the expected value of $\hat{\alpha}_1^3$, showing that it is usually biased for α_1^3 . Relate the unbiased estimator

$$\binom{n}{3}^{-1} \sum_{1 \leq i < j < k \leq n} X_i X_j X_k$$

to sample moments.

- e) Find an unbiased estimator for κ_3 based on the first three sample moments, $\hat{\alpha}_1$, $\hat{\alpha}_2$, and $\hat{\alpha}_3$.
 32. By Taylor expansion, for $\theta \in (0, 1)$,

$$-\log(1 - \theta) = \sum_{x=1}^{\infty} \frac{\theta^x}{x}.$$

From this, for $\theta \in (0, 1)$,

$$p_{\theta}(x) = \frac{\theta^x}{-x \log(1 - \theta)}, \quad x = 1, 2, \dots,$$

is the mass function for a probability distribution, called the *log series distribution*. Let X be a discrete random variable with this distribution. Find the mean and variance of X .

33. A discrete random variable X on $\{0, 1, \dots\}$ has a *power series distribution* if its mass function has form

$$P(X = x) = \frac{a(x)\theta^x}{C(\theta)}, \quad x = 0, 1, 2, \dots$$

Derive formulas for the mean and variance of this distribution involving derivatives of C .